



APPLIED PHYSICS

$^{176}\text{Lu}^+$ clock comparison at the 10^{-18} level via correlation spectroscopy

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The extreme precision of optical atomic clocks has led to an anticipated redefinition of the second by the International System of Units. Furthermore, accuracies pushing the boundary of 1 part in 10^{18} and beyond will enable new applications, such as in geodesy and tests of fundamental physics. The 1S_0 to 3D_1 optical transition in $^{176}\text{Lu}^+$ has exceptionally low sensitivity to external perturbations, making it suitable for practical clock implementations with inaccuracy at or below 10^{-18} . Here, we perform high-accuracy comparisons between two $^{176}\text{Lu}^+$ references using correlation spectroscopy. A comparison at different magnetic fields is used to obtain a quadratic Zeeman coefficient of $-4.89264(88)$ Hz/mT for the reference frequency. With a subsequent comparison at low field, we demonstrate agreement at the low 10^{-18} level, statistically limited by the averaging time of 42 hours. The evaluated uncertainty in the frequency difference is 9×10^{-19} and the lowest reported in comparing independent optical references.

INTRODUCTION

The past two decades have seen rapid development of optical atomic clocks, with many systems now reporting fractional performance at the 10^{-18} level (1–7). This performance has many important applications, such as defining the geoid (8), monitoring geopotential changes (9), measuring time and frequency (10), and testing fundamental physics (11). Independent of the proposed application, it is important to explore a wide range of atomic systems to assess their merits and long-term potential for future development. In this work, we demonstrate the ease at which two $^{176}\text{Lu}^+$ references can achieve agreement at $\lesssim 10^{-18}$. From a comparison at different magnetic fields, we obtain a quadratic Zeeman coefficient of $-4.89264(88)$ Hz/mT², which gives a corresponding fractional frequency uncertainty contribution of just 2.5×10^{-20} for comparisons at typical operating fields of 0.1 mT. A subsequent comparison with both systems at 0.1 mT demonstrates a fractional frequency difference of $[-2.0 \pm (3.7)_{\text{stat}} \pm (0.9)_{\text{sys}}] \times 10^{-18}$, where “stat” and “sys” indicate statistical and systematic uncertainty, respectively. Because of the clock transition’s insensitivity to external electromagnetic perturbations, results were obtained without magnetic shielding (12) or active field stabilization, without ground state cooling using simulations to justify its effectiveness (13), without extreme temperature measurement and control (1, 4, 5, 14), or even without the use of simulations and thermal cameras for a more moderate temperature assessment (2, 15). Such ease of use offers considerable technical advantages for practical implementations of an atomic frequency reference.

Singly ionized lutetium is a two-electron system with three available clock transitions, and the relevant level structure is shown in Fig. 1A. As described in previous work (16, 17), laser beams at 350, 622, and 895 nm facilitate optical pumping into the long-

lived 3D_1 state; laser beams at 646 nm facilitate detection, Doppler cooling, and state preparation of $|7\rangle \equiv |^3D_1, F=7, m_F=0\rangle$; and microwaves are used to drive transitions between the 3D_1 hyperfine states. This work concerns the $^1S_0 \leftrightarrow ^3D_1$ transition at 848 nm, which has favorable properties as a frequency reference. The black-body radiation (BBR) shift is $\approx -1.36(10) \times 10^{-18}$ at 300 K (18, 19), which is sufficiently small that any temperature-induced frequency difference between two references operating within a temperature range of 25° to 60°C is $\lesssim 10^{-18}$. At a typical operating magnetic field of 0.1 mT, transitions between the $m=0$ states have the lowest sensitivity to magnetic field noise of any existing clock system, being at least 5-fold smaller than those in neutral lattice clocks (1, 5), a 100-fold smaller than in Al^+ (20), and 5 orders of magnitude smaller than those in the alkaline-earth ions (Ca^+ and Sr^+) (12, 21). The large atomic mass makes it less susceptible to motional shifts than lighter ions, and the secondary clock transition at 804 nm provides exceptional detectability of micromotion. Owing to the upper-state lifetime of approximately 1 week, the dominant systematic is the probe-induced ac-Stark shift, which is 200-fold smaller than for the Yb^+ E3 clock transition under the same interrogation conditions (18, 19, 22).

RESULTS

Experimental system

All clock systems use some form of averaging, which is typically an averaging over one or more pairs of Zeeman lines (23) or over orientations of the magnetic field (24). In the case of $^{176}\text{Lu}^+$, we use an averaging over transitions from $|g\rangle \equiv |^1S_0, 7, 0\rangle$ to the hyperfine states $|F\rangle \equiv |^3D_1, F, 0\rangle$, where $F=6, 7$, and 8 (25). The average can be conveniently realized using a hybrid microwave and optical interrogation (26), which we refer to here as hyperfine-averaged Ramsey spectroscopy (HARS). As illustrated in Fig. 1 (B and C), the interrogation sequence consists of a Ramsey interrogation on the $|g\rangle \leftrightarrow |8\rangle$ transition and microwave transitions within the interrogation time to transfer population between the hyperfine states. Timing is such that the effective time in each of the hyperfine states is the same. When the laser is servoed to the central

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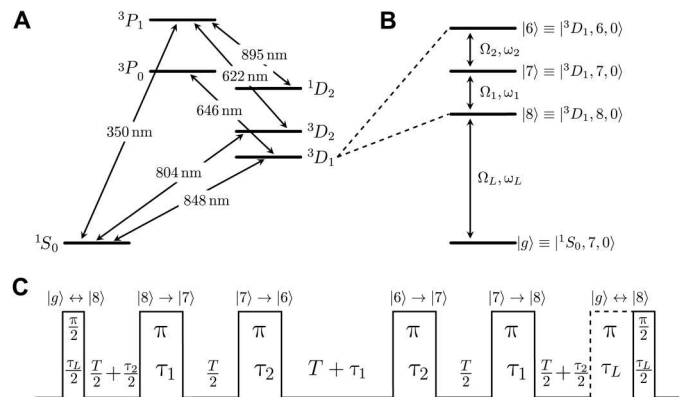


Fig. 1. $^{176}\text{Lu}^+$ spectroscopy. (A) Atomic-level structure of $^{176}\text{Lu}^+$ showing the wavelengths of transitions used. (B) Levels of the 848-nm clock transition used in the clock interrogation sequence. Ω_a and ω_a denote the coupling strengths and frequencies for the fields driving the transitions indicated. (C) Clock interrogation sequence for HARS. An optional π -pulse on the optical transition is included when implementing hyper-Ramsey spectroscopy to suppress probe-induced ac-Stark shifts.

Ramsey fringe, the hyperfine-averaged (HA) clock frequency is given by

$$f = \frac{1}{3}(f_8 + f_7 + f_6) = f_L + \frac{1}{3}(2f_1 + f_2) \quad (1)$$

where f_F is the resonant frequency of the $|g\rangle \leftrightarrow |F\rangle$ transition, $f_L = \omega_L/(2\pi)$ is the laser frequency, and $f_k = \omega_k/(2\pi)$ for $k = 1, 2$ are the microwave frequencies used within the interrogation sequence. As explained in previous work (26), the interrogation sequence ensures that $f_L + (2f_1 + f_2)/3$ represents the HA clock frequency, as indicated by Eq. 1, even if the microwave fields are not exactly resonant.

Comparisons were carried out on two ions held in separate traps, denoted Lu-1 and Lu-2, the details of which are reported elsewhere (17, 27). As our laser coherence limits interrogation times to ~ 50 ms, we use correlation spectroscopy (28–30), which allows extended probe times by factoring out laser noise. As with other implementations (30), this requires stabilization of the path length difference between the two chambers, which is achieved using standard techniques (31). In our case, we have been able to extend interrogation times to more than 700 ms, which is limited only by a heating problem in one trap (Lu-1). Each comparison consisted of three interleaved experiments: one to measure the frequency difference, δf , between the two atomic references using HARS; one to keep the 848-nm clock laser near to resonance with the $|g\rangle \leftrightarrow |8\rangle$ transition using Rabi spectroscopy; and one to monitor the magnetic field in each chamber using microwave spectroscopy of the $|^3D_1, 7, 0\rangle \leftrightarrow |^3D_1, 6, \pm 1\rangle$ transition. The duty cycle for measuring the frequency difference was 87% for the longest interrogation times used in the experiments.

Systematic effects

Typical systematic effects for the comparison experiments are given in Table 1. The dominant systematic shift arises from the probe-induced ac-Stark shift. This was measured in each trap by interleaving Rabi and hyper-Ramsey spectroscopy (22, 32) of the $|g\rangle \leftrightarrow |8\rangle$ transition, with the frequency difference determining the probe-induced ac-Stark shift, $\delta_{s,k}$ for Lu- k . For HARS, the resulting shift

given in Table 1 is (32)

$$\delta_{s,k} = \frac{\delta_k}{1 + \frac{\pi T_R}{2\tau_L}} \quad (2)$$

where

$T_R = 3(T + \tau_1 + \tau_2)$ is the total interrogation time and τ_L is the π -pulse time for the clock transition. Several ac-Stark shift measurements were taken throughout the campaign, and δ_k for each experiment is determined from measurements of nearest proximity in time. As the shift is not continuously monitored, we assumed a 5% uncertainty in the ac-Stark for each trap to allow for possible variation during the experiment but bound by the maximum variation observed between measurements. The value of δ_k given in Table 1 is a single representative example.

Excess micromotion (EMM) was evaluated on the $|^1S_0, 7, 0\rangle \leftrightarrow |^3D_2, 9, 0\rangle$ transition at 804 nm by the resolved sideband technique (33), and several measurements were carried out throughout the campaign. For each trap, the mean and half the difference of the largest and smallest determinations are given for the shift and uncertainty, respectively. Thermal motion was investigated from dephasing when driving the $|g\rangle \leftrightarrow |8\rangle$ transition and variable delays before probing were used to estimate heating rates. The associated second-order Doppler shift is dominated by a high heating rate in Lu-1, and the fractionally larger uncertainty arises from the accuracy at which that could be assessed. The heating rates were 2.3(0.8) mK/s and 0.099(15) mK/s for Lu-1 and Lu-2, respectively. These were measured repeatedly throughout the campaign, with the larger heating rate been measured almost daily.

Time-varying magnetic fields give rise to an ac-Zeeman shift, with the dominant contributions coming from currents in the electrodes driven by the radio frequency (RF)-trapping potential. The contribution depends on the component of the RF-magnetic field perpendicular to the applied dc field (34). This was measured from an Autler-Townes splitting on the Ba^+ clock transition (27), giving root mean square (RMS) values of 0.98(1) μT and 0.52(1) μT for Lu-1 and Lu-2, respectively.

The microwave fields used in HARS can also contribute to an ac-Zeeman shift (26). Inasmuch as practically possible, microwave antennas are positioned to maximize the π component and balance the $\sigma^\pm$ components, which suppresses the contribution to $\lesssim 7 \times 10^{-19}$ for any of the comparisons reported here. The microwave field components are deduced from measured π times on $|7, 0\rangle \leftrightarrow |F, m_F = 0, \pm 1\rangle$, where $F = 6, 8$ and the uncertainty given in the table is based on a 5% inaccuracy for the measured π times. Detailed calculations are given in the Supplementary Materials. The hyperfine interaction gives rise to a quadrupole shift that is not eliminated by HA. The shift is characterized by a residual quadrupole moment of $-2.5 \times 10^{-4} ea_0^2$ as determined from a combination of g -factor measurements and theory (16). The shift can then be estimated using the quadrupole shift extracted from $S_{1/2} \leftrightarrow D_{5/2}$ clock measurements in Ba^+ and rescaling the quadrupole moments. This has been done for Lu-1 (27), which gives an estimated shift of 2.2×10^{-19} with a 10% uncertainty ascribed to the value of the quadrupole moment. For Lu-2, the maximum possible shift from the dc confinement would be 3.2×10^{-19} if the magnetic field was aligned along the trap axis. For both traps, the magnetic field is at an angle of approximately 57° to the trap axis. Hence, we would expect a reasonable suppression of the component from the trapping potentials

Table 1. Characteristic uncertainty budget. All values are relative to 10^{-18} of the HA frequency. The values given here are for experiment 7. Interrogation parameters are $\tau_L = 8$ ms, $\tau_1 = \tau_2 = 12$ ms, $T = 200$ ms, $B_1 = 0.1$ mT, and $B_2 = 0.1$ mT. Tables for experiments 1 to 6 are given in the Supplementary Materials.

Effect	Lu-1		Lu-2		Difference	
	Shift	Unc.	Shift	Unc.	Shift	Unc.
EMM	−0.41	0.37	−0.44	0.34	0.03	0.50
Second-order Doppler (thermal)	−1.87	0.45	−0.13	0.06	−1.75	0.45
ac-Zeeman (RF)	0.54	0.01	0.15	0.01	0.39	0.01
ac-Zeeman (microwave)	−0.06	0.04	−0.06	0.15	−0.01	0.16
Gravity shift*	—	—	—	—	−1.31	0.15
ac-Stark shift	−126.59	6.33	−125.14	6.26	−1.45	8.90
HARS timing	2.60	1.33	4.88	0.72	−2.28	1.51
Microwave coupling	0	0.21	0	0.21	0	0.30
Optical coupling	0	0.28	0	0.28	0	0.39
Residual quadruple shift	0.22	0.02	0	0.32	0.22	0.32
Total shift	−125.58	6.50	−120.74	6.33	−6.15	9.07

*Only differential shifts are considered.

and the shift dominated by stray fields. As it is unlikely that a shift from stray fields would exceed the maximum possible shift from the dc confinement, we bound the shift by $\pm 3.2 \times 10^{-19}$.

In the analysis of HARS (26), it was assumed that the detuning of the microwaves was much smaller than the associated coupling strengths, and the finite duration of the optical pulses was neglected. For the level of precision obtained here, these assumptions should be reconsidered. A comprehensive analysis is given in the Supplementary Materials, and the relevant findings are summarized in Materials and Methods. The finite duration of the optical pulse is associated with an effective time $2\tau_L/\pi$ spent in [8]. As this was not accounted for by reducing the dwell time in [8], it results in an effective timing error. In addition, there is a much smaller correction that is practically independent of the optical pulse time. However, the influence of this term depends on the microwave and optical coupling strengths relative to the values determined by their respective π times. The uncertainties given in Table 1 for the coupling errors correspond to 1 and 2.5% uncertainty in the microwave and optical couplings, respectively.

A differential gravitational shift of $1.31(15) \times 10^{-18}$ is included to account for a height difference of 1.2(1) cm between the ions. All other shifts are expected to contribute negligibly to the frequency difference compared to the statistical uncertainty achieved in the comparisons. A discussion of these omitted shifts is given in Materials and Methods.

Measurement of quadratic Zeeman coefficient

The quadratic Zeeman coefficient (α_z) of the $|^1S_0\rangle \leftrightarrow |^3D_1\rangle$ transition can be determined by

$$\alpha_z = \frac{\delta_z}{B_1^2 - B_2^2} \quad (3)$$

where B_k is the applied static magnetic field for Lu- k and δ_z is the frequency difference between the two atomic references appropriately corrected for all other systematic shifts. The results from six measurements are given in Fig. 2, offset by the weighted mean. From all six measurements, we obtain $\alpha_z = -4.89264(88)$ Hz/mT² with $\chi^2_\nu = 1.30$, as discussed in Materials and Methods. With $B_1 = 1.9$ mT and $B_2 = 0.1$ mT, the estimated uncertainty in α_z requires agreement of the two atomic references to within 9.0×10^{-18} when held at the same magnetic field. A further comparison (experiment 7) was carried out with both systems at 0.1 mT to confirm this, which gave the result $[9.2 \pm (3.6)_{\text{stat}} \pm (9.1)_{\text{sys}}] \times 10^{-18}$, as determined by four separate measurements with a total integration time of 29.3 hours. As indicated in Table 1, the comparison is limited by the assessment of the probe-induced ac-Stark shift. Nevertheless, the measured frequency difference is consistent with and supports the assessed uncertainty in α_z . At a typical operating field of 0.1 mT, the reported uncertainty in α_z contributes 2.5×10^{-20} to the fractional frequency uncertainty.

Comparison using hyper-Ramsey

Probe-induced ac-Stark shifts can be heavily suppressed using hyper-Ramsey spectroscopy (32, 35, 36), and the principle also applies to correlation spectroscopy and HARS. Following the work of Yudin *et al.* (22, 32), a hyper-Ramsey pulse can be included, as indicated in Fig. 1C, with the clock laser frequency shifted by the measured ac-Stark shift during the clock pulses, and the additional hyper-Ramsey π -pulse phase shifted by π with respect to the first $\pi/2$ -pulse. As shown in the Supplementary Materials, this not only suppresses the ac-Stark shift but also eliminates the effective timing error from the finite optical π time and the sensitivity to the optical coupling strength.

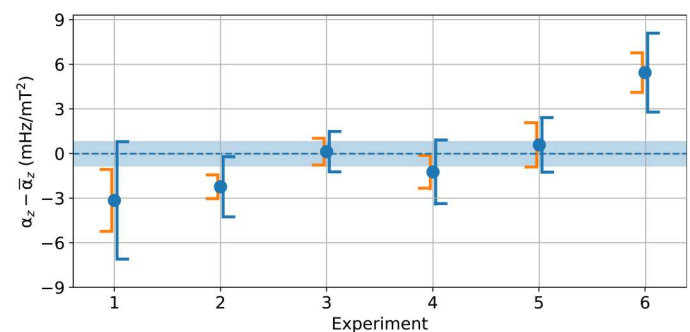


Fig. 2. Measurements of the quadratic Zeeman coefficient (α_z). Uncertainty budgets and parameters τ_L , τ_1 , τ_2 , T and static applied fields B_1 and B_2 for each experiment are given in the Supplementary Materials. Each point is the mean of all data collected at the given parameters, corrected for systematic errors, and offset by the weighted mean of all measurements. Error bars on the left represent the statistical uncertainty, and those on the right are inclusive of the systematic uncertainty. The shaded region indicates the uncertainty in the weighted mean. From all six measurements, we obtain the estimate $\alpha_z = -4.89264(88)$ Hz/mT², with $\chi^2_\nu = 1.30$.

A final comparison (experiment 8) including a hyper-Ramsey pulse gave $[-2.0 \pm (3.7)_{\text{stat}} \pm (0.9)_{\text{sys}}] \times 10^{-18}$, which is inclusive of all systematics given in Table 2, as determined by 10 measurements with a total integration time of 42.4 hours. Data are shown in Fig. 3, and an error budget is given in Table 2. With suppression of the probe-induced ac-Stark shift, the only shifts above 10^{-18} are the SD shift and the gravitational shift, and these are still less than the measurement precision. Moreover, the SD shift is primarily due to a heating problem in one trap. Consequently, clock agreement at this level is not reliant on substantial corrections and control of large systematics as it is for all other clock comparisons that have been made thus far.

DISCUSSION

In summary, we have used correlation spectroscopy to compare two $^{176}\text{Lu}^+$ atomic references with an inaccuracy at the low 10^{-18} level. This has provided a high accuracy assessment of the quadratic Zeeman shift, which is the leading systematic for the $^{176}\text{Lu}^+ 1S_0 \leftrightarrow 3D_1$ clock transition. Hyper-Ramsey spectroscopy is also compatible with HARS and practically eliminates the probe-induced ac-Stark shift as the next leading systematic. This then provides a frequency reference with systematics that are easily controlled to an inaccuracy below 10^{-18} without the technical challenges faced by other systems.

An accuracy claim for an individual system would require an assessment of the BBR shift. Presumably, the use of simulations and thermal cameras (2, 15) could equally be used here, although it is unclear how to quantify such an assessment for our system. A crude analysis as given in Materials and Methods would bind the temperature to $35(10)^{\circ}\text{C}$, which, due to the low BBR shift for the $1S_0 \leftrightarrow 3D_1$ transition, corresponds to a BBR shift of just $-1.56(27) \times 10^{-18}$ and gives a total uncertainty for both systems of $<8 \times 10^{-19}$. However, although it is done elsewhere (2, 37), we refrain from making any accuracy claims without a same-species comparison

to the appropriate level. Here, stability limits us to the low 10^{-18} , which is primarily limited by a heating problem in Lu-1.

With the lower heating rate and real-time monitoring of EMM, systematics at the mid- 10^{-19} are readily achievable. Correlation spectroscopy with a 10-s interrogation time was demonstrated in our previous work (38), albeit within the same trap. In separate chambers, uncorrelated magnetic field noise may be a limiting factor. However, at the operating field of 0.1 mT, the maximum magnetic sensitivity of the 848-nm clock states is $\sim 0.4 \text{ Hz}/\mu\text{T}$. This is two orders of magnitude smaller than Al^+ , for example, for which lifetime-limited correlation spectroscopy has been claimed (30). Increasing the interrogation time to an anticipated 10 s would make comparisons below 10^{-18} achievable on the time scale of a few days.

The relative ease at which a comparison at this level can be achieved with this system should not be underrated. Comparisons between clock systems claiming this level of accuracy (39, 40) have variations of the measured ratios that cannot be explained by the error budgets of the individual systems, which is compelling evidence that large systematic shifts are not being controlled to the levels claimed. Such problems are unlikely to occur for systems such as lutetium or thulium (41) for the simple fact that systematics are not large enough to cause such variations. The total shift and any single contribution given in Table 2 is, at most, at the level of the current measurement precision. Moreover, it is self-evident that systems having such low systematics have substantial room for improvement if current state-of-the-art technologies were used to take

Table 2. Uncertainty budget for comparison using hyper-Ramsey (experiment 8). All values are relative to 10^{-18} of the HA frequency. Parameters τ_L , τ_1 , τ_2 , T , B_1 , and B_2 are as for experiment 7, in Table 1.

Effect	Lu-1		Lu-2		Difference	
	Shift	Unc.	Shift	Unc.	Shift	Unc.
EMM	-0.41	0.37	-0.44	0.34	0.03	0.50
Second-order Doppler (thermal)	-1.87	0.45	-0.13	0.06	-1.75	0.45
ac-Zeeman (RF)	0.54	0.01	0.15	0.01	0.39	0.01
ac-Zeeman (microwave)	-0.06	0.03	-0.13	0.11	0.07	0.11
Gravity shift*	-	-	-	-	-1.31	0.15
Microwave coupling	0	0.21	0	0.21	0	0.30
Residual quadruple shift	0.22	0.02	0	0.32	0.22	0.32
Total shift	-1.58	0.68	-0.55	0.60	-2.34	0.93

*Only differential shifts are considered.

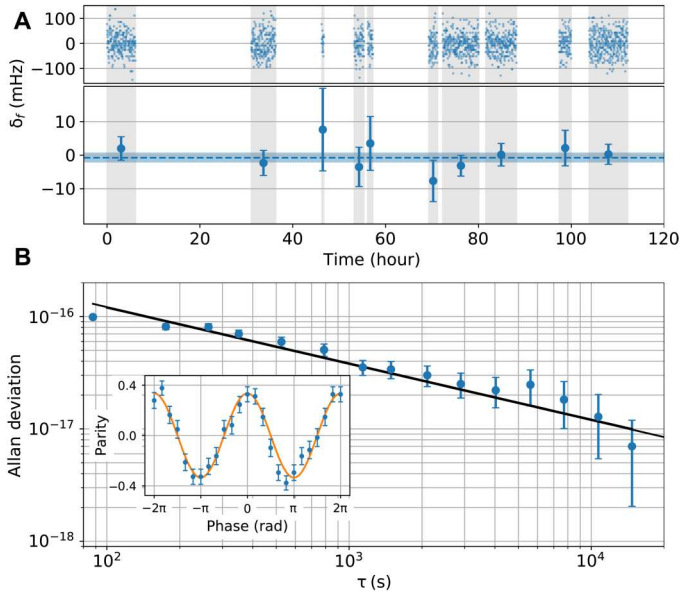


Fig. 3. Measurement instability. Comparison data between two $^{176}\text{Lu}^+$ references implemented via HARS, including hyper-Ramsey for ac-Stark shift suppression (experiment 8). **(A)** Frequency difference measured over 42.4 hours in 10 runs of the experiment. Points are the mean of each run, and error bars are the statistical error determined by the projection noise limit. The dashed horizontal line is the weighted mean ($\chi^2_\nu = 0.45$), and the shaded region is the corresponding uncertainty, which represents a fractional frequency difference of $[-2.0 \pm (3.7)_{\text{stat}} \pm (0.9)_{\text{sys}}] \times 10^{-18}$. **(B)** Fractional Allan deviation for the longest single run. The inset shows a typical parity signal as a function of the relative phase of the second $\pi/2$ -pulse, the amplitude of which is limited by heating in Lu-1.

advantage of this potential. This is particularly noteworthy for lutetium as the magnetic field sensitivity of the clock states is the lowest of any atomic system that we are aware of and the lifetime is practically indefinite. Consequently, interrogation times will not be limited by lutetium in any foreseeable future.

MATERIALS AND METHODS

Additional clock shifts

Clock shifts considered but not included in the error budget are given here. These shifts are unlikely to affect the comparisons made in this work.

ac-Zeeman shifts

Additional ac-Zeeman shifts arise from two sources. The first concerns the ac coupling between Zeeman states. These do not cause a clock shift directly, but they do modify the effective g factor of the level or equivalently the B field determined by measuring a Zeeman splitting. With the RF field at the microtesla level, this only affects the measured field at the 10^{-9} level. At 50 Hz, we estimate field noise to be around 8 nT RMS, which similarly affects the measured field at the 10^{-9} level. The second source arises from the coupling between fine-structure levels. This has both a scalar and tensor contribution, but the latter is eliminated by HA. The former is determined by $\alpha_z \langle B^2 \rangle$, where B is the total ac field. This has the same magnitude as the measured perpendicular component, as determined by measurements from three orthogonal directions (17). For determination of α_z , this influences $\langle B_1^2 \rangle - \langle B_2^2 \rangle$ by $\lesssim 10^{-6}$ fractionally compared to the 1.7×10^{-4} fractional inaccuracy of α_z . Similarly, the influence on a comparison when the clocks are both at low field is $\sim 10^{-20}$.

The BBR shift

The polarizability for the 848-nm transition in $^{176}\text{Lu}^+$ has been very well characterized (18, 19). The corresponding BBR shift is the lowest of any clock candidate under active investigation with an estimated shift of $-1.36(10) \times 10^{-18}$ at 300 K. For the comparison result reported, we are only concerned with the frequency difference. Even allowing for one trap at 25°C and the other 45°C, the contribution to the frequency difference would be just $\sim 5 \times 10^{-19}$ and well below our measurement resolution. Given the near identical construction of the two traps and the similar trap drive powers supplied to each, such a temperature differential is highly unlikely. We also note that a temperature assessment with an inaccuracy of 2°C has been reported (2), which was based on measurements made on a trap of identical construction, with simulations to account for different mounting within the vacuum chambers (15). In our case, the mounting is also identical. In such an assessment, it must be assumed that traps of near identical construction have near identical temperatures, at least better than 2°C, in fact. Although we do not suggest that our traps have a temperature difference of $\lesssim 2^\circ\text{C}$, we think it reasonable to assume that the temperature difference is less than 20°C and does not have any relevance to the comparisons made here. As we are unable to make a measurement of the temperature in either system and any reasonable temperature difference is immaterial to the final result, we do not include this in the comparison assessment.

Timing accuracy

For HARS, timing accuracy of the pulse sequence can influence the dwell time in each state, which could potentially degrade the averaging. If the dwell time for each F is $T + \tau_1 + \tau_2 + \delta T_F$, then, using equation 7 from (26), the clock shift δ will be

$$\delta = \frac{1}{T_R} \sum_F \Delta_F \delta T_F = \frac{\Delta_2 \delta T_6 - \Delta_1 \delta T_8}{T_R}$$

where Δ_F are linear combinations of the microwave detunings Δ_1, Δ_2 as defined in (26). For the second equality, we have used the fact that $\langle \delta T_F \rangle = 0$ as any nonzero value can be absorbed in the definition of T . The on-board 100-MHz clocks of the digital Input/Output boards are externally synchronized, so the main considerations are switching delays of clock laser acousto-optic modulator (AOMs) and microwave sources. Switching delays of the clock laser AOMs are ~ 60 ns. However, as this does not influence the overall dwell time in $|8\rangle$, it does not result in a clock shift. Switching time differences of microwave sources only influence the synchronization at the 5-ns level. Hence, for the hertz level detunings of the microwave fields, fractional shifts are well below 10^{-21} for all experiments reported here.

AOM chirp

The switching of AOMs and microwave sources can potentially introduce artificial frequency shifts (42) even when path lengths are nominally stabilized. Following the methodology reported in (43), this was investigated in preliminary experiments and estimated to be below 10^{-19} for the experiments reported here.

Imperfections in the implementation of HARS

As noted here, microwave detunings, the finite duration of the optical pulses, and imperfect coupling strengths of the microwave and optical fields can compromise the effectiveness of our interrogation technique. A detailed analysis of these effects is provided as a supplement, but we summarize the main findings here. The leading order effect is due to the finite duration of the optical pulses, which is associated with an effective time $2\tau_L/\pi$ spent in $|8\rangle$. When not accounted for, it results in an effective timing error given by eq. S6. An additional error given by eq. S20 has the same form, which arises from deviations of the optical coupling Ω_L from the ideal condition $\Omega_L \tau_L = \pi$. From the analysis following eq. S22, both of these corrections are practically eliminated when including a hyper-Ramsey pulse. As seen in eq. S16, there is an additional term that is practically independent of τ_L , and this term is very well approximated by the analytic expression given in eq. S18, which is an oscillatory function with an amplitude that is quadratic in the microwave detunings normalized by the coupling strengths. For the microwave detunings used here, this term has no significant contribution to the analysis and is only slightly modified when including a hyper-Ramsey pulse. However, a contribution from this term arises from imperfect microwave couplings as discussed in relation to eq. S21. As illustrated in figs. S5 and S6, the effect is to add a constant to the quadratic amplitude, and we use this amplitude to bind the contribution as an uncertainty.

Microwave detunings used in this work were measured and set with a resolution of approximately 0.1 Hz. However, at the larger operating field of 1.9 mT, variations in the magnetic field could change the detunings by up to ± 2 Hz. Data analysis has included

these variations using the corrections provided by eq. S6. As a consistency check, eq. S18 was also used to confirm that it gave no statistically significant difference in the final results.

Uncertainty in α_z

Complete measurement results and uncertainty budgets for experiments 1 to 6 are given in the Supplementary Materials. The leading systematic is due to ac-Stark shift, which is uncorrelated as it is re-measured between experiments. For this reason, the statistical and ac-Stark uncertainties are summed in quadrature, and only these are used to evaluate the weighted mean value $\alpha_z = -4.89264(81)$ with $\chi_\nu = 1.3$. The HARS timing systematic is correlated across experiments as all microwave detunings are dependent on just a few parameters via eq. S7 and S8. These parameters are randomly resampled within their stated uncertainties to find the additional variance in the weighted mean α_z . The SD is added in quadrature to uncertainty in α_z above to arrive at $-4.89264(88)$. All other systematics are generally correlated across experiments but do not contribute significantly to the uncertainty.

Probe-induced ac-Stark shift

For the final comparison, which includes a hyper-Ramsey pulse, we have omitted any contribution from the probe-induced ac-Stark shift. Similar to experiment 7, the probe-induced ac-Stark shift was approximately 7 Hz during the pulses with a differential between the two systems of 0.52 (0.47) Hz. As evidenced by experiment 7, the effect on the comparison is already at the level of the measurement precision even without the hyper-Ramsey pulse. Consequently, any level of suppression would be beyond our measurement precision. As shown in the Supplementary Materials, for small microwave detunings, the shift has the same cubic dependence on the effective ac-Stark shift with only minor modifications from the microwave detunings. Hence, even if the frequency was not shifted during the optical pulses, the shift would only influence the comparison at the mid- 10^{-19} level. As the net effective shift is trivially stable at the 0.5 Hz level, the shift of each system will be limited by the microwave coupling accuracy, which is already accounted for in the error budgets.

Supplementary Materials

This PDF file includes:

Supplementary Text
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References

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$^{176}\text{Lu}^+$ clock comparison at the 10^{-18} level via correlation spectroscopy

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